David Ludington

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Education

Loyola University Chicago Chicago, IL
Bachelor of Science in Computer Science, Magna Cum Laude *August 2021 – May 2025*

Technical Skills

Languages: Python, TypeScript
Frameworks: Playwright, Pytest, unittest, Selenium, Appium, DeepEval
Test Automation: UI, API, end-to-end, regression
Specialized Testing: geo-consent testing, accessibility testing
CI/CD: Github Actions, Git

Experience

QA Automation Engineer | ***Encyclopedia Britannica*** May 2024 – Present

- Built UI test automation frameworks from the ground up for a consumer-facing game platform serving 2 million monthly users, leveraging Python, Selenium, and Appium to automate testing across desktop, iOS, and Android devices.
- Orchestrated AI-assisted engineering workflows using GitHub Copilot, developing custom agent instructions and skills to coordinate multi-agent task execution, optimize context utilization, and streamline test development and debugging.
- Reduced existing automated test execution time by 75%, from 3 hours to 45 minutes, by migrating legacy unittest suites to Pytest, using parallel execution with fixtures and parameterization to improve maintainability.
- Enabled earlier regression detection by integrating automated test suites into GitHub Actions workflows triggered by pull requests, scheduled runs, and production deployments.

Writing Center Tutor | ***Loyola University Chicago*** August 2022 – May 2025

- Provided personalized feedback to over 200 students, helping them strengthen their writing skills and improve their academic work.
- Conducted workshops on academic writing strategies, improving student outcomes across various disciplines.

Fitness Supervisor Lead | ***Loyola University Chicago*** February 2023 – May 2025

- Managed a team of 50 fitness supervisors, optimizing communication channels and operational workflows.
- Created a comprehensive scheduling framework that balanced staff availability with facility demands, reducing scheduling conflicts.
- Ensured compliance with safety protocols, contributing to a safe environment.

Research

Test Effectiveness | *Python, Git* February 2024 – June 2024

- Extracted and analyzed software metrics to assess test coverage and effectiveness.
- Authored technical documentation summarizing findings, contributing to improved testing practices and better software quality.

Doyle Living Systems Laboratory | *Docker* May 2021 – May 2022

- Designed and maintained virtual environments using Docker, streamlining workflows for the laboratory team.
- Developed expertise in interpreting technical documentation and managing container systems.

Projects

Schedule Builder for Campus Recreation | *Python, Pandas* May 2024

- Automated scheduling for a team of 50 employees by creating a tool that aligns employee availability with facility operating hours.
- Enhanced scheduling efficiency, reducing manual input errors and freeing up time for other critical tasks.

Snake Game | *C++, Raylib* February 2024

- Designed and implemented a fully functional Snake game using C++ and Raylib, exploring advanced graphics capabilities.

Image to ASCII art | *Python, tkinter* March 2024

- Developed a Python script to convert JPEG images into ASCII art.